

Water Level Alarming with Water Pump Control

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Abstract— Efficient water management is a critical requirement in residential and industrial settings to prevent water wastage and maintain uninterrupted supply. This project presents a cost-effective and reliable water level monitoring and control system using a 555 Timer IC and relay-based pump control. The system continuously senses the water level in a tank and activates an LED indicator when the water reaches a predefined high or low level. Simultaneously it controls a water pump to either stop or start water flow based on the detected level. The circuit utilizes simple electronic components including conductive probes, a 555 Timer configured in bistable mode, a transistor driver and a relay mechanism to switch the pump. This design not only automates the water refilling process but also minimizes manual intervention and potential overflow or dry-run damage. Its simplicity, low cost and ease of implementation make it ideal for household and small-scale industrial applications.

Keywords— 555 Timer, Water Level Control, Relay, bistable mode.

I. INTRODUCTION

Manual control of water pump systems is inefficient and often results in water overflow, pump damage or dry-run conditions. To address these issues, this project introduces a simple water level alarming and pump control circuit based on the 555 Timer IC in bistable mode. The circuit automates the switching of a water pump and provides visual alerts using an LED when specific water levels are reached. The 555 Timer, when configured in bistable mode functions as a flip-flop that retains its output state until triggered by an external signal making it ideal for ON/OFF control applications such as pump automation [1]. Analog timer-based circuits are widely used in basic control systems due to their reliability, low power consumption, and ease of implementation without requiring microcontrollers [2]. This system uses water-sensing probes to detect level thresholds in a storage tank. When a probe is activated, the circuit changes state and drives a relay through a transistor, which in turn operates the water pump. An LED indicator is used for visual feedback. The circuit is low-cost, easy to build and highly effective for small-scale water management applications.

II. LITERATURE REVIEW

Automation in water level control has gained attention due to its practicality in resource management and pump protection. Several studies have explored simple yet effective methods using basic analog electronics to eliminate manual intervention in tank refilling systems. The 555 Timer IC is widely used in ON/OFF control circuits due to its flexibility in monostable and bistable configurations. Research has shown that timer-based circuits provide a cost-effective solution for water level management without requiring complex programming or microcontrollers [3]. Another study demonstrated the successful use of bistable circuits to implement logic-level switching with reliable latching behavior, which is essential for maintaining the motor's ON or OFF state based on input triggers [4]. Additionally, diode-protected relay drivers have been highlighted as essential in safely interfacing low-power control circuits with high-power AC loads like water pumps [5]. These findings support the choice of using a 555 Timer IC and relay-based approach in the presented project. The selected method offers simplicity, low cost, and dependable performance suitable for domestic and educational applications.

III. METHODOLOGY

The design of the water level alarming and pump control system is based on a simple and reliable circuit using a 555 Timer IC configured in bistable mode. The circuit monitors the water level using conductive probes placed at specific heights within a water tank. When the water touches a probe, it provides a low-resistance path to trigger the timer circuit, which in turn activates or deactivates the water pump through a relay. An LED indicator is used to visually signal the status of the system. The entire setup is built using basic components such as resistors, transistors, a relay and a 555 Timer IC, ensuring low cost and ease of implementation.

A. Circuit Diagram:

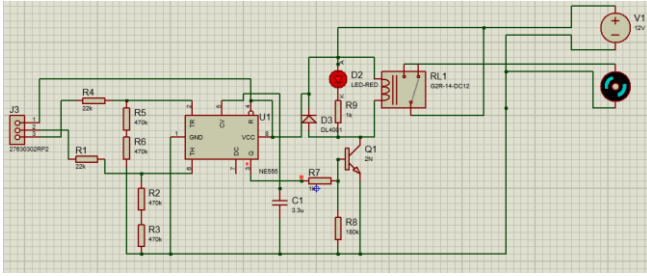


Fig-01: Water Level Detection and Automatic Pump Switching Circuit

B. Components:

i. 555 Timer IC:

The 555 Timer IC is configured in bistable mode to act as a flip-flop. It switches its output state based on the trigger inputs from water level probes. When a probe detects water contact, the timer either sets or resets the output, which ultimately controls the pump via a relay. This IC provides stable switching performance without the need for microcontrollers or complex logic.

ii. Transistor:

The transistor acts as a switching device. When the 555 Timer output goes high, it provides base current to the transistor, turning it on. The transistor then allows the current to flow through the relay coil, activating the relay to power the pump.

iii. Diode:

The diode is connected in parallel with the relay coil to protect the transistor from voltage spikes generated by the collapsing magnetic field when the relay turns off. This is known as a flyback diode and is essential for safe transistor operation.

iv. Relay:

The relay functions as an electromechanical switch that allows the low-power control circuit to operate a high-power AC pump. It is triggered by the transistor and physically connects or disconnects the pump from the power supply based on the water level.

v. Capacitor:

A capacitor is included for noise filtering or debounce purposes in the trigger circuit, preventing false triggering of the 555 Timer. It smooths out any fluctuations in the input voltage from the probes, improving the stability of the system.

vi. LED:

The LED provides a visual indication of the system status. It lights up when the water reaches a certain level and the pump is turned on, giving immediate feedback to the user about the operation of the system.

vii. Probes:

The probes act as level sensors and are placed at specific heights in the tank. When water touches a probe, it completes a low-resistance path to ground, triggering the 555 Timer. Different probe levels can be used for detecting LOW, MID, or High-water levels as needed.

C. Working Principle:

The circuit operates based on two water level probes placed at predefined positions inside the tank—one at the LOW level and the other at the HIGH level. These probes send input signals to a 555 Timer IC configured in bistable mode, which maintains its output state until a trigger signal changes it. The timer output controls a transistor-relay combination that switches the pump ON or OFF, along with an LED for visual status.

1) Pump Turn on:

When the water level falls below the Low-level probe, it indicates that the tank is nearly empty. This condition triggers the SET pin of the 555 Timer, changing its output to HIGH. The high output signal turns on the NPN transistor, which completes the circuit for the relay coil, energizing it. As a result, the relay switches on the water pump, starting the refilling process. Simultaneously, the high output also powers the LED, providing a visual indication that the pump is currently running.

2) Pump Turn off:

As the tank refills, the water level gradually rises and eventually meets the High-level probe. This triggers the RESET pin of the 555 Timer, changing its output to LOW. The low output cuts off the base current to the transistor, turning it off. This de-energizes the relay coil, disconnecting the pump from the power supply, and stopping the water flow. The LED also turns off, indicating that the water tank has reached the desired level and the pump is no longer active.

D. Bread-board Implementation:

The entire circuit was first assembled and tested on a breadboard to verify its functionality and ensure proper triggering of the 555 Timer and relay operation based on water level detection.

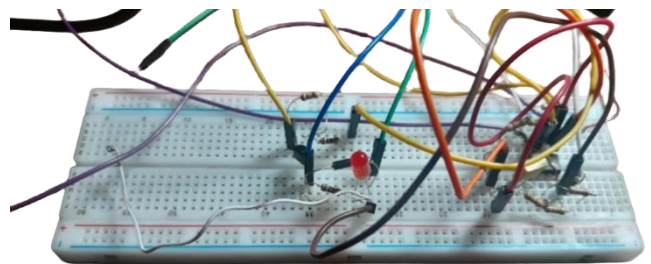


Fig-02: Bread-board implementation

E. PCB Design:

The circuit was also simulated in Proteus to observe real-time behavior before hardware assembly, and a custom PCB layout was designed for a more permanent and compact implementation.

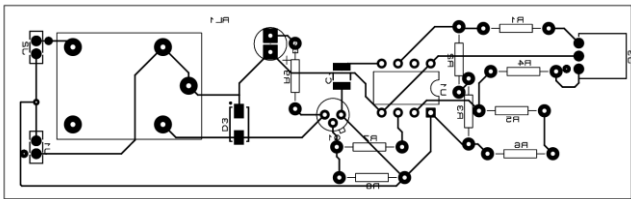


Fig-03: PCB Layout

F. Hardware Implementation:

The final version of the circuit was implemented on a custom PCB, which was fabricated using the toner transfer method and etched with ferric chloride solution. After etching, all components were carefully soldered onto the board, and the system was tested to ensure proper functionality under real conditions.

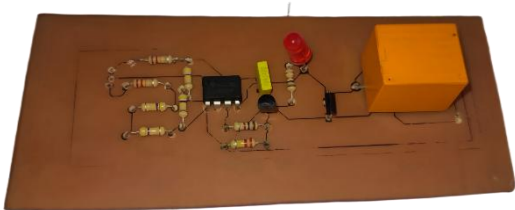


Fig-04: Final hardware setup

IV. RESULT

The circuit performed accurately in controlling the water pump based on the detected water levels. When the water dropped below the lower probe, the motor was automatically turned ON, and the LED indicator lit up, signaling that the pump was in operation. As the tank filled and the water reached the upper probe, the motor switched OFF instantly, and the LED also turned off, indicating that the water level had reached the maximum threshold. The transitions between ON and OFF states were smooth, without any flickering or false triggering. The LED provided clear visual feedback throughout the process. The relay operated reliably, and the switch between pump states was fast and stable. The system maintained its state properly until the next valid water level change occurred. This consistent performance confirmed the effectiveness of the 555 Timer in a bistable mode for such ON/OFF control tasks. The overall result demonstrates that the circuit is well-suited for practical, real-world water tank automation, offering both reliability and simplicity.

V. COST ANALYSIS

Component	No. of component	Price (BDT)
555 Timer IC	01	30
Resistor	09	9
Transistor	01	8
Capacitor	01	4
Relay	01	50
Diode	01	3
LED	01	45
Breadboard	01	180
Connecting Wires Set	01	30
Copper Board	01	20
Ferric Chloride	20 gm	10
Battery	01	300
	Total	689

This project was completed with a total cost of only 689 BDT, making it highly affordable and accessible for students and hobbyists. All components, including the breadboard, 555 Timer IC, relay, LED, and other basic parts, were sourced locally at minimal cost. Despite its low budget, the circuit delivered reliable and consistent performance.

VI. APPLICATIONS AND FUTURE WORK

This water level alarming and pump control system can be applied in various real-life scenarios such as household water tanks, agricultural irrigation, industrial fluid tanks, and overhead water reservoirs. Its simplicity and low cost make it suitable for use in rural areas or small-scale installations where full automation systems are impractical.

In future developments, the system can be enhanced by integrating a buzzer for audible alert, or by using multiple probe levels for more detailed water level indication. It can also be upgraded with microcontroller-based logic to support remote monitoring, mobile notifications, or smart home integration for a more advanced and feature-rich system.

VII. DRAWBACKS

Although the system performs reliably, it has a few limitations:

- It uses a manual probe-based method, which may degrade over time due to corrosion or mineral deposits.
- The design is limited to two level detections (low and high), which restricts more granular control.
- The system lacks real-time digital display or data logging features.
- Power failure during operation may cause the 555 Timer to lose its output state.

VIII. CONCLUSION

This project successfully demonstrates a low-cost, efficient, and reliable solution for automatic water level detection and pump control. By using a 555 Timer IC in bistable mode along with basic components, the system automates the tank refilling process and provides a visual indicator of motor operation. It significantly reduces manual effort, prevents water wastage, and protects the pump from dry-running. The simplicity of the design makes it suitable for academic, domestic, and small-scale industrial applications.

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IX. REFERENCES

- [1] M. M. Abrar, "Design and Implementation of Astable Multivibrator using 555 Timer," IOSR Journal of Electrical and Electronics Engineering (IOSR-JEEE), vol. 12, no. 1, pp. 22–29, 2017. DOI: 10.9790/1676-1201022229
- [2] D. Vidyasagar, "Real Time Simulation of Working of Timer IC 555 using its Internal Block Diagram," Shri R.L.T. College of Science, 2019. DOI: 10.13140/RG.2.2.11385.01124
- [3] R. M. Walia, "Automatic Water Level Controller using Timer IC," International Journal of Innovative Research in Electrical, Electronics, Instrumentation and Control Engineering, vol. 4, no. 6, pp. 25–28, 2016.
- [4] A. Jaiswal and A. Singh, "Design of Bistable Multivibrator Using IC 555 for Logic Control Applications," International Journal of Engineering Trends and Technology (IJETT), vol. 67, no. 3, pp. 53–56, 2019.
- [5] S. Rajput and K. Sharma, "Relay Interfacing with Electronic Control Circuits: Design Considerations," Journal of Electronics and Control Systems, vol. 8, no. 1, pp. 14–19, 2020.